

Timing issues of constant bit rate services over ATM

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One of the problems associated with ATM is the variable transport delay introduced by the cell switches and through waiting time jitter which can perturb any constant bit rate stream traversing the network. This paper introduces the three techniques that are available to counter such cell delay variation (CDV) and reconstitute constancy of the bit stream exiting the network. Each of the three techniques discussed varies in the amount of success achievable in eliminating jitter and wander from the exiting bit stream. Of particular importance is to map the capabilities and economics of each technique to the service being offered such that each service achieves adequate bit rate constancy, consistent with that service's quality of service objectives.

1. Introduction

When transporting constant bit rate streams over an asynchronous transfer mode (ATM) network it is necessary to ensure that both the transmitter and receiver clocks are properly synchronised. On reception the received data, encapsulated in ATM cells, would normally be loaded into a play-out buffer at the full physical line rate, in cell bursts, before being clocked out bit-by-bit by the receiver's clock at a lower service rate. Two degradation mechanisms come into play when the transmitter and receivers' clocks are not synchronised. Firstly, if the receiver clock is offset from the transmitter or if cells arrive too early or too late from the ATM network, the play-out buffer will overflow or underflow resulting in either lost bits or the need for padding bits. Secondly, the use of an inappropriate synchronisation method can result in jitter and wander of the clock associated with the constant bit rate (CBR) data outbound from the ATM network. Such clock imperfection can lead to observable defects on the end service such as 'wow' and 'flutter' in hi-fi acoustic services, visible imperfections such as colour impairments on video services, bit errors due to alignment jitter when interworking with the plesiochronous (PDH) or synchronous digital hierarchy (SDH) network or frame slips when interworking with existing PSTN and narrowband ISDN (N-ISDN) networks. This paper briefly outlines the techniques available for clock synchronisation between transmitter and receiver to ensure that synchronisation imperfections caused by cell delay variation (CDV) are within acceptable bounds.

2. Cell delay variation

When a constant bit rate stream is fed into an ATM network the stream is mapped into the ATM cell payload and the ATM cells passed to the physical layer at regular intervals related to the time to fill the ATM payload at the service clock rate.

For example, for a 2.048 Mbit/s CBR service utilising ATM adaptation layer-1 (AAL-type 1) with 47 bytes available in the payload, ATM cells are passed to the physical layer every $47 \times 8 / 2.048 \text{ Mbit/s} = 183.6 \text{ } \mu\text{s}$.

(It should be noted that, although ATM generally offers a 48 byte user payload, AAL-type 1 makes use of one payload byte for the cell sequence number, cell sequence number protection and convergence sub-layer indication — see Fig 7 in section 3 for more details.)

On traversing the ATM network, in the absence of cell delay variation and with the receiving service clock locked to the transmitting service clock, the cells would arrive with constant inter-interval time, the 47 bytes would be loaded into a 47 byte buffer and the data clocked out bit-by-bit by the service clock. Just as the buffer is about to empty the next cell would arrive to replenish the buffer (Fig 1).

However, if each cell is subjected to a different delay through the network and, for example, if the second cell arrives late, the buffer would empty and have no more bits to send. If the second cell arrives early, the 47-byte buffer would overflow. Therefore, even with matched transmitter receiver clocks the play-out buffer needs to be dimensioned and operated in such a way as to allow for late or early cell arrivals. The usual technique is to delay playout of the initial cell by the calculated maximum cell delay variation allowance. This has the effect of ensuring no buffer depletion even when the first cell arrives very early and the second cell arrives very late (Fig 2(a)). However, the buffer now needs to be enlarged to at least the payload size plus the number of cells that could be delivered in an interval equal to twice peak-to-peak CDV. This ensures that, if the first cell actually arrives very late and is additionally

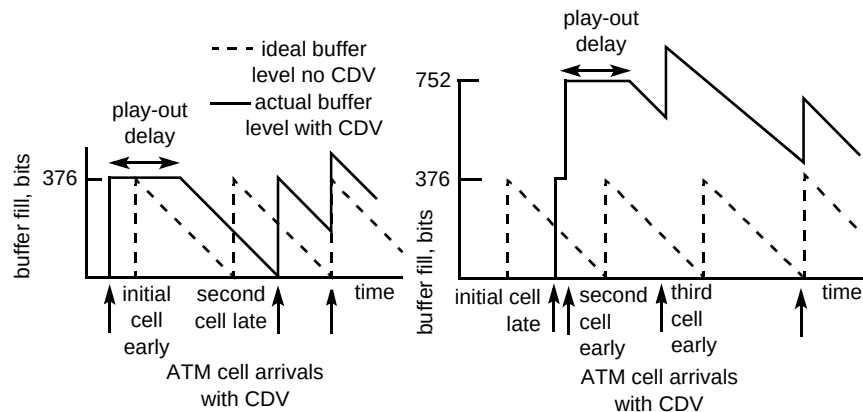


Fig 1 Playout buffer.

delayed by the peak-to-peak CDV before playout and subsequent cells arrive very early, there will still be space available for them in the buffer (Fig 2(b)). In this latter case the buffer level needs to be optimised over a period of time to ensure the cells are delayed in the buffer for a period not in excess of the peak-to-peak cell delay variation.

Where a receiver's clock takes time to lock to the transmitter, or where the receive clock varies in a time-dependent fashion, the buffer level will vary beyond that influenced by CDV and the time taken to lock and the frequency difference over time should be taken into account when specifying the delay needed before initial playout and total buffer size. The delay in the buffer can be thought of as the elimination of variable delay by an increase in total fixed delay — the total fixed delay of the ATM network thus becomes the fixed transmission propagation delay plus the peak-to-peak CDV.

The main mechanism, among many, responsible for cell delay variation within an ATM network is output port contention where cells destined for the same output port of an ATM switch arrive simultaneously on different input ports. Whilst one cell is played out on the physical line other cells have to be loaded into a buffer to await their turn. Given independent traffic streams on each input port (and given an output-port buffered, non-blocking, switch design)

cells arriving at the switch at a regular rate will have to wait varying amounts of time in the output buffer dependent on the number of cells instantaneously contending for the same output port. The regularity of the input traffic stream is thus disrupted and cells arrive at their destination at irregular intervals. Other mechanisms that can contribute to CDV include that of waiting time jitter where the intercell period dictated by the transmitting service clock does not map exactly to both the cell interval available on the physical transmission line, and the variation in delay experienced through the switching fabric of an ATM switch.

The main parameter affecting buffer dimensioning at the receiver and initial play-out delay is thus peak-to-peak cell delay variation. However, as will be seen later in the paper, when using an adaptive clock recovery synchronisation technique the time dependence of CDV can adversely affect the purity of the recovered clock leading to larger playout buffer dimensions (or, in the absence of a larger playout buffer, cell loss), and jitter and wander of the output bit stream.

Many published papers model the characteristics of cell delay variation for various network reference models, network loads and traffic characteristics. Many make use of analytical queuing models, such as $N^*/D/D/1$ or $M/D/1$ models for a single switch, as a basis for network

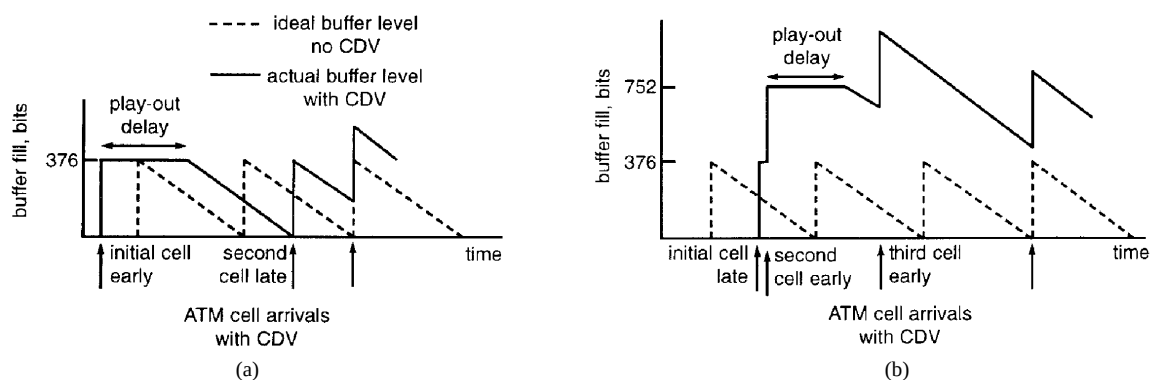


Fig 2 Playout buffer dimensioning to absorb CDV.

evaluation, others use numerical simulation techniques, and yet others attempt to load prototype ATM switches with 'real' traffic in an attempt to characterise the CDV likely to be incurred in a 'real' network. No one technique can exactly predict the performance expected of real networks since the network's characteristics will be dynamic and strongly influenced by the network operator's traffic management policies. For example, such a policy may dictate the traffic loads allowed by connection admission control (CAC), the service partitions, traffic shaping functions and the service-dependent buffer management algorithms to be enacted for different services and differing qualities of service. For networks with switch output ports lightly loaded, little or no CDV would be incurred, whereas in networks where constant bit rate traffic has to contend with large and varying switch loads, large cell delay variation can be expected. For each specific case, the peak-to-peak CDV incurred by the service and its time-dependent characteristics will be very different. The onus is therefore on each network operator to ensure that its traffic management policy and simulation studies evolve towards a network design capable of meeting the requirements of each specific service that it offers on its ATM network.

For simulations attempting a worst case analysis, the M/D/1 queuing model, which assumes Poisson arrivals and deterministic service time, is viewed by many as adequate for predicting worst case CDV.

As an example the waiting time distribution of an M/D/1 queue is given by

$$Q(x) = \sum_{n=x}^{\infty} \frac{(\rho(n-x))^n}{n!} e^{-\rho(n-x)} (1-\rho)$$

and for $\rho = 0.9$ (i.e a 90%) switch load the peak-to-peak CDV (defined here as the 10^{-3} quantile for comparison with Fig 3a which comprises ~ 1000 samples) is calculated as $x = 34$ cell times. For an STM-1 output port this equates to a peak-to-peak CDV of 96 μs .

Such pessimistic results can be used to engineer robust buffer management policies. However, such analytical techniques give little information on the expected time-

dependence of CDV since by default the models imply an essentially 'white' frequency spectrum for a static load. Numerical simulation techniques and 'real' measurements therefore provide the best opportunity to understand the frequency characteristics of CDV. The cell delay characteristics and CDV power spectral density, resulting from a simulation of 2 Mbit/s CBR connection multiplexed with 15 other Bernoulli traffic sources for a fixed total output buffer load of 90%, is shown in Figs 3(a) and 3(b).

The limited results of the simple numerical model support the analytical model results in that an essentially 'white' power spectral density is predicted. However, caution has to be applied to such simplistic modelling. Models such as these again, generally, tend to evaluate the effects of a static load on the output buffer. Any slow changes in network load over the day could result in strong spectral lines at low frequencies. Any step changes in network load, contention of CBR streams with bursty variable bit rate streams or dissimilar rate CBR streams, or the effects of waiting time jitter can also result in strong spectral characteristics at both low and high frequencies. As an example, Figs 4(a) and 4(b) show the effect of a step change in network load from 70% to 90% on the power spectral density. Low frequency components are now seen to dominate.

It is important therefore that not only are the cell level effects at static load understood but equally important is an understanding of the effects of network load statistics. Such effects can restrict adaptive clock recovery techniques to only selected CBR services in which the higher levels of jitter and wander can be tolerated.

3. Eliminating or reducing the effects of CDV on timing

Three solutions exist at the ATM layer to eliminate the effects of CDV from constant bit rate connections; these comprise:

- fully network-synchronous operation by using a network derived clock as the service clock,

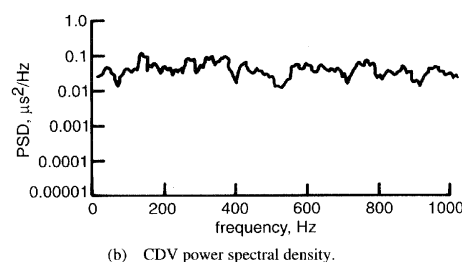
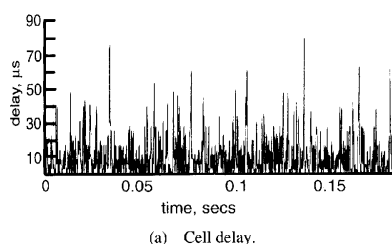


Fig 3 Simulation of 2 Mbit/s CBR for a 90% buffer load.

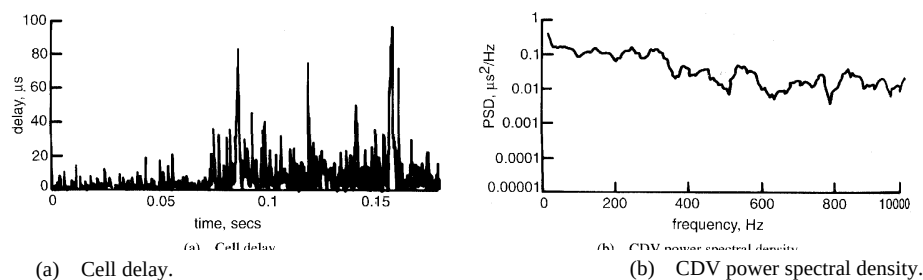


Fig 4 Simulation of 2 Mbit/s CBR with step change in network load.

- the use of synchronous residual time stamps (SRTS) to transmit service clock information,
- adaptive clock recovery.

Additionally, where the network is solely responsible for transporting ATM cells, rather than delivering a constant bit stream to the user, the receiving higher level application layer can have the added responsibility for service clock derivation.

3.1 Network-synchronous operation

In forcing an end station to operate in synchronism with the network clock the effects of CDV can be completely eliminated. As shown in Fig 5 the service clock is derived directly from the network clock. Given the presence of a network clock at both the transmitting and receiving end-stations, the service bit stream can simply be clocked out of the playout buffer at the service bit rate, fully independent of the cell delay variations caused by the ATM network. A

playout buffer with correct dimensions and initial delay in playout is still required to ensure that CDV does not cause cell loss nor padding bits to be required. In the event of network clock failure, the buffer will also require either a buffer slip algorithm with the receive clock placed in hold-over mode or the receiver must revert to an adaptive clocking algorithm. The network clock is distributed through the synchronisation of the physical layer connecting the ATM network elements. These ATM network elements may themselves be synchronised via an external physical timing interface or as a slave of another ATM network element via the physical layer connection.

3.2 Synchronous residual time stamp

The network synchronous option described in section 3.1 has the drawback that not all end stations and CBR sources would wish to synchronise to the network clock. In these instances one option is to transport information about the source clock across the ATM network to eliminate playout buffer depletion or overflow. The technique

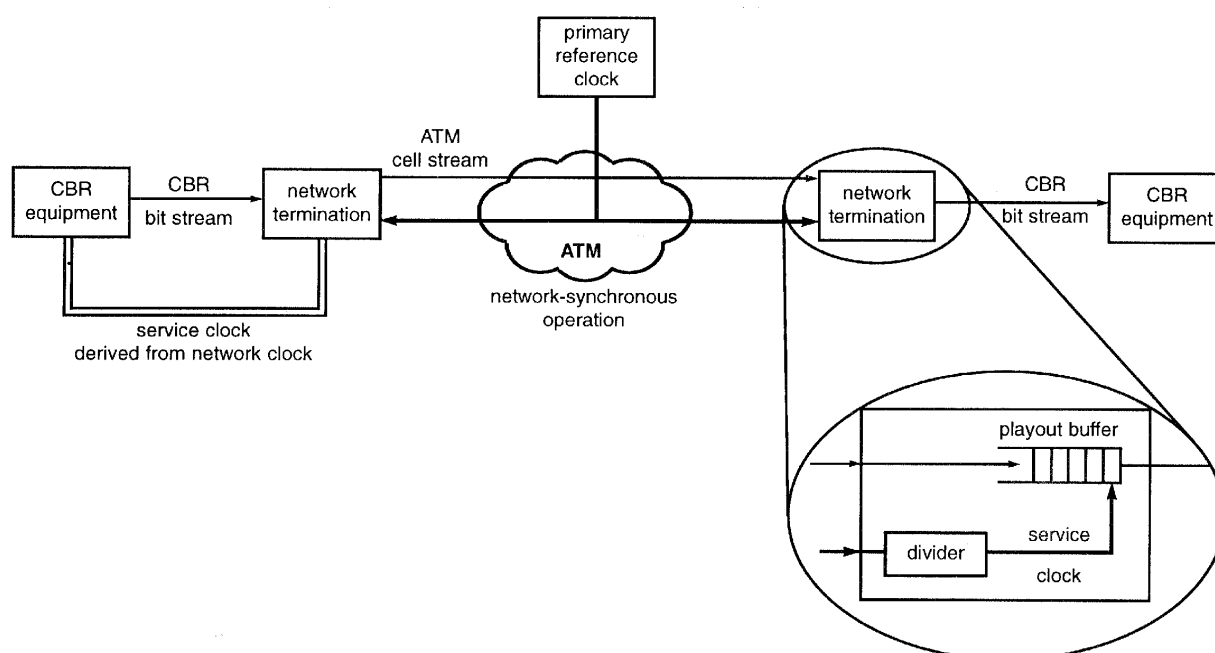


Fig 5 CBR connection using a fully networked synchronous operation.

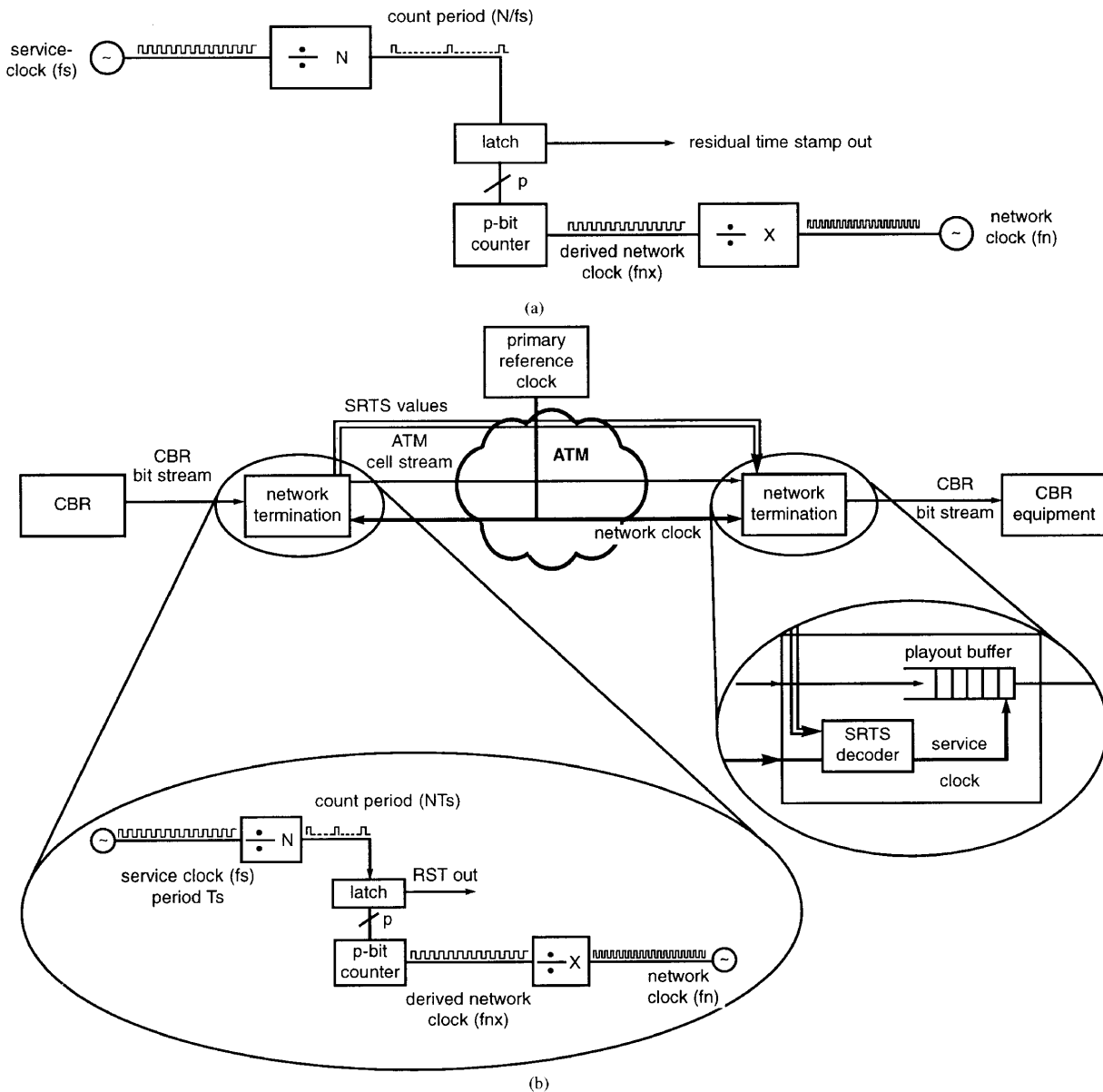


Fig 6 An implementation of the SRTS synchronisation technique.

adopted for this is called the synchronous residual time stamp (SRTS) technique which is closely associated with the ATM adaptation layer type 1 protocol. To allow SRTS to work accurately the network needs to distribute a good quality network clock to the network interface.

Its general implementation is shown pictorially in Figs 6(a) and 6(b).

With SRTS the number of derived network clock (fn/X) cycles in N cycles of service clock is counted in a p -bit counter and this p -bit SRTS value transmitted across the network to the receiving network terminating point. At the receiving terminal the local service clock can be adjusted

against the reference network clock to ensure that the same number of derived network clock cycles are counted over the same N cycles of local service clock, thus locking the local service clock to the transmitter service clock by use of the received residual time stamps and the reference network clock.

For ITU G.702, 1.544 Mbit/s and 2.048 Mbit/s-based CBR hierarchies the value of $N = 3008$ and a $p = 4$ -bit SRTS value has been agreed and specified in ITU-T draft recommendation I.363.x.

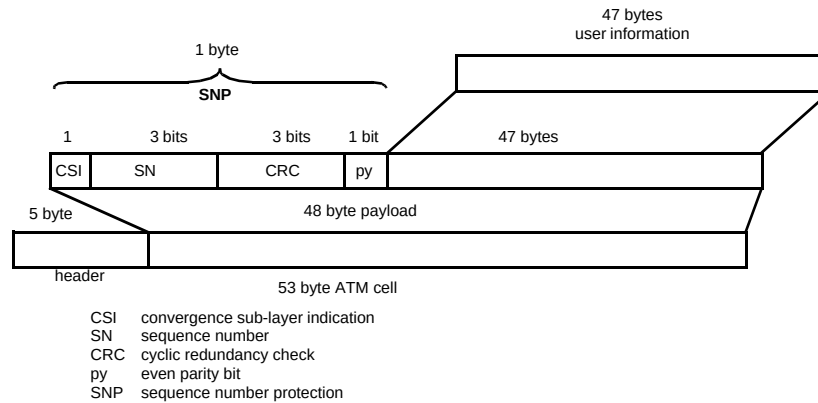


Fig 7 AAL type 1 — non-structured data format.

The derived network clock f_{nx} is specified uniquely by assuming the existence of an 8kHz clock derived from the network frame rate and constraining f_{nx} by the following:

$$f_{nx} = 8 \text{ kHz} * 19440 / 2^k \text{ where } k = 0, 1, 2, \dots, 11$$

$$1 \leq f_{nx} / (f_s) \leq 2$$

The 4-bit SRTS value is transported over the ATM network as convergence sub-layer indication (CSI) bits distributed alternately over eight consecutive ATM cells using ATM adaptation layer type 1 (Fig 7).

The value of $N = 3008$ has been chosen which corresponds to the number of AAL-type 1 user information bits in eight ATM cells.

Since the nominal service clock frequency is known by both transmitter and receiver the full derived network clock count need not be sent. Hence, the use of only 4 bits as a 'residual', or off-set, rather than full count. This residual count gives an indication of offset from nominal clock frequency. The use of 4 bits allows up to a ± 200 ppm offset from nominal clock frequency.

The quantised nature of the residual time stamp will lead to slight clock imperfections at the receiver. The effect of quantisation may be visualised by examining the implementation details for a specific service such as a 2.048 Mbit/s CBR service. In this particular implementation the derived

network clock is specified, from the constraints given above, as 2.43 MHz. The number of derived network clock cycles in 3008 service clock cycles is counted in 4-bit modulo-16 counter. For a 10 ppm offset service frequency the total count would be 3569.0268 derived network clock cycles. The 4-bit residual time stamp truncates this to 3569 and sends the residual modulo-16 count of 0001 binary. With knowledge of the nominal service frequency the truncated derived-network-clock count of 3569 can then be generated at the far end on reception of the RTS value 0001 binary. The service clock generated from this truncated value is given by $2.43\text{MHz} * (3008/3569) = f_s = 2.048 \text{ MHz} + 17.5 \text{ ppm}$, i.e. some 7.5 ppm in error due to the RTS quantisation. This static frequency error, and thus increasing phase error, will continue until the partial clock counts of 0.0268 cycles per RTS period accumulate (over $1/0.0268 = 37$ RTS periods) to greater than unity giving a 3570 clock count and implied service clock frequency for this RTS value of $f_s = 2.048 \text{ MHz} - 262.6 \text{ ppm}$, this time -272.6 ppm in error from the transmit service clock. However, in this case over the long term of 37 RTS periods the average frequency information given $= (37 * 17.5 \text{ ppm}) - 262.6 \text{ ppm} = +10.128 \text{ ppm}$, which begins to approach the true transmit service frequency. Thus, by taking the longer term average of the RTS values, the true underlying transmitted service frequency can be determined and tracked. The overall effect is best viewed by looking at the time-dependent phase difference of the received service clock implied by the RTS values to the actual transmit clock. This is shown in Fig 8(a)

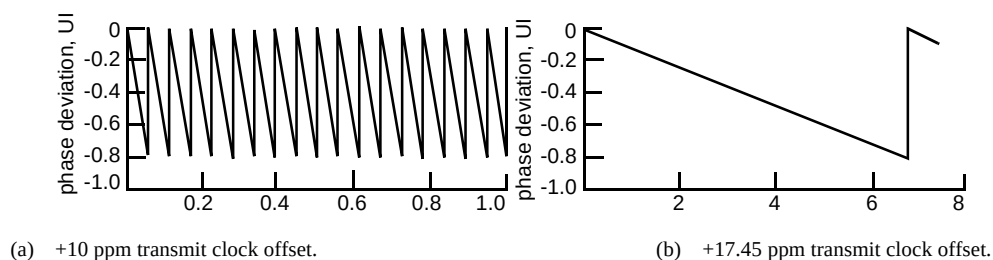


Fig 8 RTS phase transients.

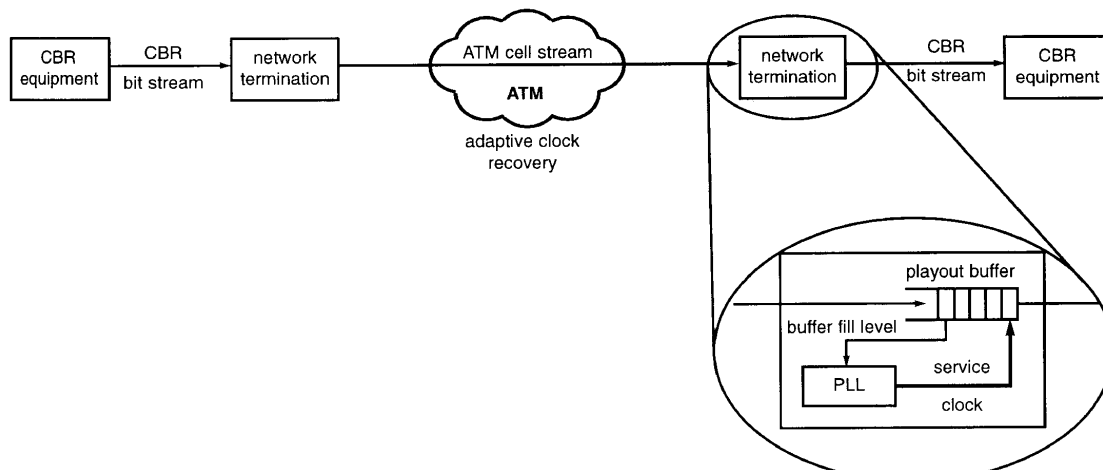


Fig 9 An implementation of adaptive clock recovery.

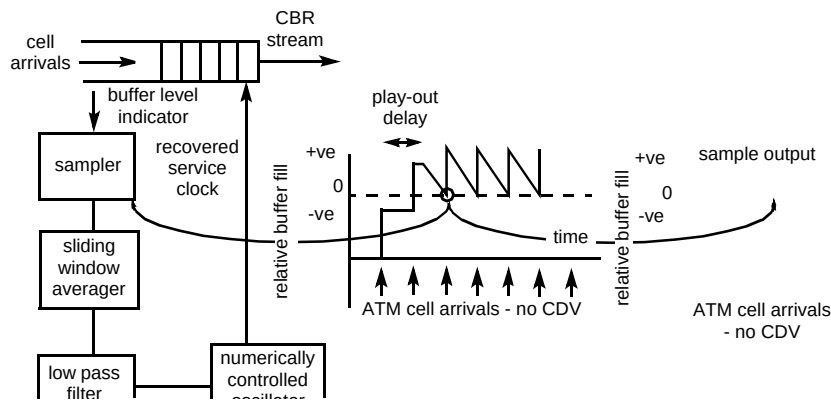


Fig 10 Principle of adaptive clock recovery.

for the example just given. The induced phase transients in this particular case are sufficiently frequent that if necessary they could be reduced by a sufficiently long time constant in the phase-locked loop being used to derive the receive service clock.

For certain values of service frequency, particularly where the derived network count is close to an integral number, the phase transients will become so infrequent that it becomes more difficult to filter them out, as shown in Fig 8(b). However, even in this case the magnitude of the isolated phase transients fall inside the ITU-T G.823 recommendation limits for wander and jitter on PDH interfaces. Therefore, provided a good quality network clock is delivered to the network interfaces, the SRTS technique is capable of delivering third-party timing across the ATM network within the jitter and wander constraints imposed by ITU-T G.823.

3.3 Adaptive clock recovery

Where ATM cells traverse a number of different ATM networks, it may not be possible to provide a common

network clock to the transmitting and receiving end-stations. Also, some services may not require the purity of clock available from the above network synchronous or SRTS schemes. In these instances it may only be possible or more cost effective to use an adaptive clock recovery technique.

The technique of adaptive clock recovery is depicted in Figs 9 and 10 which serve as an introduction to the basic theme.

In general, adaptive clock recovery relies on the fact that, irrespective of the amount of delay variation experienced in the network, the CBR cell stream has an underlying average inter-arrival time. The job of any adaptive clock recovery mechanism is to extract this longer term average inter-arrival time from the 'noise' produced by both CDV and cell loss, and to use this as a basis for received service clock derivation.

The data buffer acts as a phase comparator of the incoming cell stream and outgoing bit stream — the buffer

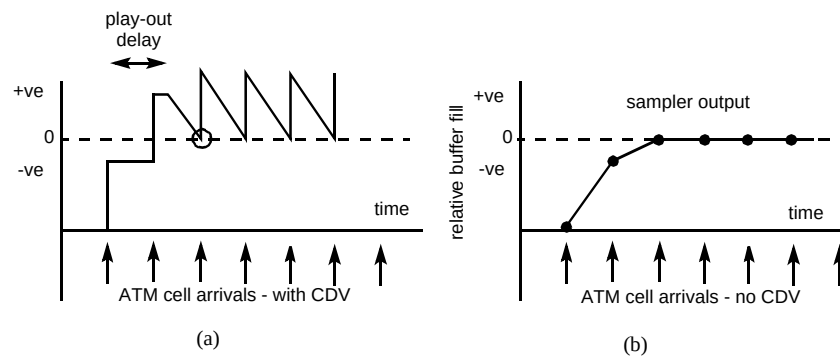


Fig 11 Clock difference causing increasing buffer level.

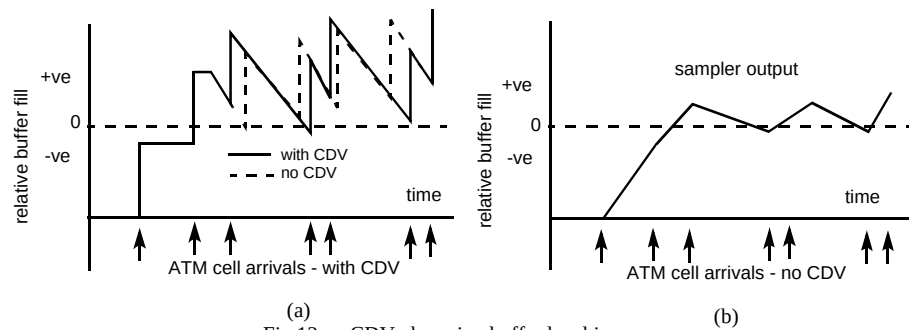


Fig 12 CDV obscuring buffer level increase.

fill level being the output of the phase comparison. Since on cell arrival the service bits are delivered 'in bulk' by the ATM payload, the buffer fill varies in a 'saw-tooth' fashion as in Fig 10(b), even in the absence of CDV.

If this waveform is sampled at the cell arrival rate, in the absence of CDV and/or waiting time jitter, and with the transmitting and receiving clocks locked the sampled values will be of constant value as in Fig 10(c).

If the receiving clock is too slow compared with the transmitter, the average buffer level will rise with time and the sample values would gradually increase in value as in Fig 11.

This trend, once detected, can be used to increase the receive clock frequency to bring the buffer level back to a nominal value and lock in the receiving clock to that of the transmitter. Unfortunately, the presence of CDV can obscure the underlying trend as shown in Fig 12.

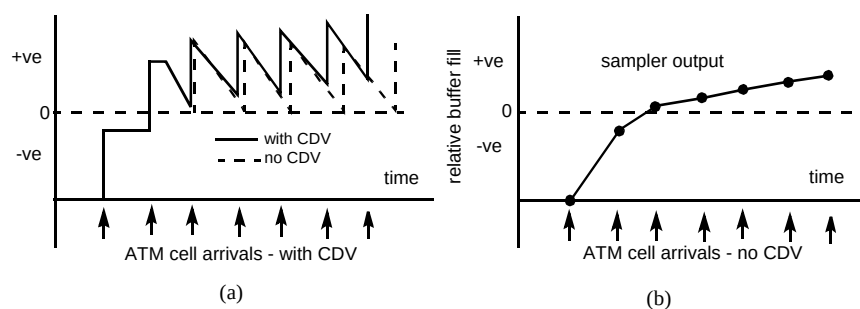


Fig 13 Low frequency CDV causing increasing buffer level.

The buffer level sample values cannot therefore be used directly but must be filtered in such a way as to allow the recovered clock to react only to a mismatch of the transmitter to receiver clocks. In general, such a mismatch would result in a continued increase or decrease in average buffer fill level over the averaging period. The adaptive recovery algorithm should ignore the relatively more rapid fluctuations around the longer term average due to CDV.

The adaptive clock recovery scheme therefore acts as a low-pass filter. As a result low-frequency CDV, as may occur due to a slowly cyclic or step change in network load, will be misinterpreted as a frequency mismatch, as shown in Fig 13. With low frequency CDV the buffer level will increase or decrease over the averaging period of the adaptive algorithm in much the same way as that due to transmitter-receiver mismatch. The adaptive recovery circuit will erroneously respond to low frequency CDV producing low frequency phase variation of the recovered clock in concert with the CDV.

To demonstrate this, the effect of the CDV characteristic of Fig 4(a) (and reproduced over a longer time period in Fig

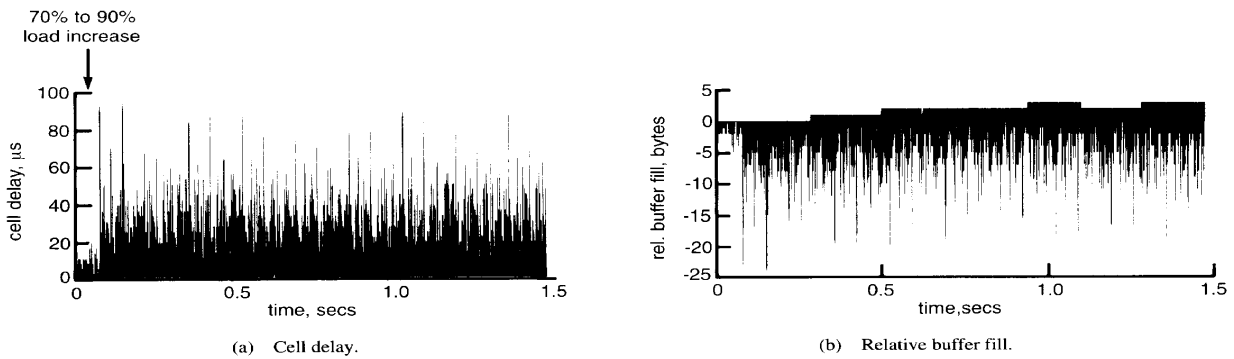


Fig 14 Effects of CDV and local change on buffer fill.

14(a)) on a simulated 2.048 MHz adaptive clock recovery scheme is shown below in Figs 14(a) to 15(c).

The late cell arrivals in Fig 14(a) are reflected as low buffer levels in Fig 14(b).

The output frequency deviation, total output phase deviation (wander) and jitter above 20Hz is shown in Figs 15(a) to 15(c) respectively.

The adaptive clock recovery mechanism can be seen to respond to the rms increase in cell delay when the switch load instantaneously increases from 70% to 90%. The rms increase in delay causes a general lowering of the buffer level and the output frequency is decreased momentarily to bring the mean buffer level back to normality. The frequency transient in this case causes a peak-to-peak jitter above 20 Hz of 0.5 unit intervals (UI) (Fig 15(c)) and a total output phase deviation equivalent to the increase in rms transport delay of 24.4 UI = 11.9 μs (Fig 15(b)). These results are for a single switch; in general, the load characteristics for all switches through which the CBR cell stream passes must be taken into account by the simulation, and the peak-to-peak phase deviations, due to network load variations, will easily exceed the values shown here.

When considering the use of adaptive clock recovery for any specific service the following characteristics need to be accounted for in the design:

- the transmitting service clock parameters including frequency, tolerance and slew rate,
- the CDV temporal and spectral characteristics which will be imposed on the service cell stream by the ATM network(s) it traverses — including the effects of port contention and waiting time jitter,
- the phase transfer function associated with the adaptive clock recovery mechanism,
- the jitter/wander tolerance of the specific service being carried.

The adaptive clocking mechanism needs to be able to track any changes of the transmitting service clock within its given bounds of tolerance and slew rate, whilst at the same time not tracking changes implied erroneously by CDV within the network. By default CDV will impart some fluctuations to the recovered clock dependent upon the magnitude of the CDV within the network and the CDV-to-received-service clock jitter/wander transfer function of the

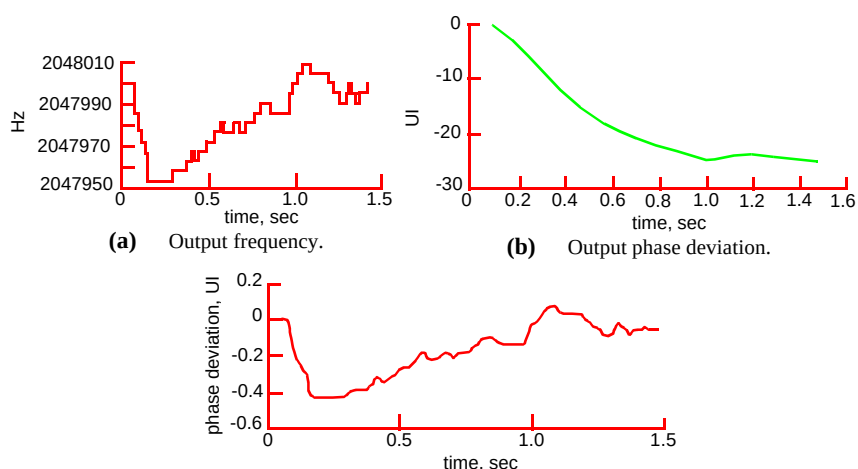


Fig 15 Effect of network load transient on receiver clock.

adaptive clocking mechanism. However, the design must ensure that the ensuing phase variations are within the bounds of the service specific jitter/wander tolerance specifications.

4. Service requirements

Currently, the services that dominate the telecommunications market are the PSTN and N-ISDN circuit-switched services, KiloStream and MegaStream type constant bit rate bearer services and the packet switched X.25, frame relay and SMDS data services.

As these services are migrated from their individual switching platforms to a common ATM broadband ISDN (B-ISDN) switching platform, it is clear that interworking issues between existing switching platforms will be a key issue. In addition, the ATM platform will have to support an almost limitless number of new B-ISDN services, for example video-on-demand, broadcast TV, interactive conversational videotelephony services, interactive computer games, and multimedia conferencing. Such services may exist entirely in the ATM domain, being incompatible with existing switch platforms, while others, such as interactive videotelephony, will have to interact with existing switch platforms such as the N-ISDN videotelephony service.

It is essential therefore when investigating network synchronisation requirements for an ATM B-ISDN switch platform that both interworking issues with existing platforms and the requirements for services contained entirely within the ATM domain be considered. Service-specific tolerances and choices for service-specific synchronisation strategies are now discussed.

4.1 PSTN and N-ISDN services — transport and interworking

Existing PSTN and N-ISDN switch platforms operate in a fully network-synchronous fashion at 2.048 Mbit/s and have digital switch designs which reflect the expected characteristics of the physical transmission links which interconnect them. One key aspect is the need for long term frame synchronisation between switch input ports. Although the interconnecting transmission links have virtually fixed delay, these links exhibit low frequency delay variation caused by day-to-day ambient temperature changes. This delay, although small, requires some additional buffering within the switch to minimise frame slip events. As a guide, ITU Recommendation G.823 specifies a tolerance limit related to this diurnal variation of just 18 μ s. In the ATM domain, transmission delays are no longer fixed and day-to-day, hour-to-hour transmission delay variation could easily exceed the G.823 wander limits. As an example, in the simple model earlier (Fig 3a) the mean network delay increases by 12 μ s for one switch

alone for a load increase of just 20%. As the network load slowly varies through busy hours and periods of inactivity, any adaptive clock recovery scheme can do no more than follow such long-term variation. It is clear, therefore, in considering this one aspect alone that adaptive clocking techniques should not be used when transporting PSTN or N-ISDN services over ATM switch platforms. Instead, network synchronous operation of the customer premises equipment with that of the PSTN or N-ISDN network platform to which it wishes to interwork would ensure long-term frame synchronisation. One method of accomplishing this is to synchronise the transmission frames of the ATM platform to a primary reference clock which is common to the PSTN or N-ISDN network platform. The primary reference clock will then be traceable to the customer premises equipment via the physical layer attaching the customer to the ATM network.

4.2 PDH CBR transport

When considering the point-to-point transport of ITU-T G.702 PDH constant bit rate streams over ATM networks, the jitter and wander limits defined by ITU-T G.823 need to be met. However, ITU-T G.823 does not formally define the PDH interface output limits for longer term phase variation (wander) below 20 Hz. A casual observer may therefore be misled into thinking that adaptive clocking mechanisms could be used when transporting ITU-T G.702 CBR streams across an ATM network — particularly where the adaptive clock synchronisation mechanism can be shown to restrict the jitter imparted to PDH CBR stream to be within the G.823 output jitter limits specification above 20 Hz. However, caution has to be applied to such speculation. Although not specifying the network interface output wander limits, ITU-T G.823 does assume that the lower frequency phase variation will be limited by the physical nature of the transmission links. This was before the advent of ATM. In making this assumption ITU-T G.823 advises that, provided downstream equipment can tolerate wander limits commensurate with the fundamental propagation characteristics of transmission media and ageing of clock circuitry, then frame slips associated with the downstream equipment will be rare. As discussed above, when using adaptive clock recovery, the imparted wander characteristic, is likely to be in excess of the equipment tolerance limits specified by ITU-T G.823. Proper functioning of the equipment being interconnected by G.702 constant bit rate streams would therefore be at the mercy of the ATM network load characteristics and that equipment's own wander margin to the ITU-T G.823 recommendation. Additionally, dependent upon the transient behaviour of the adaptive clock recovery circuit and the relationship of PDH service bit rate to the ATM physical layer transport rate, it is possible to exceed the >20 Hz output jitter limit at the PDH interface. Dependent upon the specific PDH service bit rate to ATM physical layer rates, waiting time jitter or even transient loads at the ATM

switches could cause such excursions with consequential timing misalignment and bit errors at the decision circuit of the downstream PDH receiver. For these reasons the use of the synchronous residual time stamp technique is preferred to ensure both error-free operation of downstream PDH interfaces and compliance with the downstream equipment ITU-T G.823 wander tolerance specification.

It should be noted that, when using synchronous operation for PSTN/ISDN interworking or SRTS for PDH transport, the ATM platform will need to be provided with an interface to obtain a feed from a primary reference clock. Many ATM switch manufacturers already provide this, but not all.

4.3 *Pure ATM-based services*

For services that reside entirely within the ATM transport and switching domain and which do not have to interact with the existing PSTN, N-ISDN switching platforms, nor make use of existing PDH bearer services, the constraints imposed by ITU-T G.823 do not apply. Here the use of adaptive clock recovery is entirely feasible. However, consideration need now be given to the specific application and the jitter/wander tolerance requirements at ATM adaptation layer and application layer. Many new ATM-based services such as MPEG video distribution are currently being considered for use with the adaptive clock recovery technique, and for each application the merits and limitations of adaptive clock recovery, network conditions and application requirements will need to be reviewed in detail to ensure a successful robust implementation.

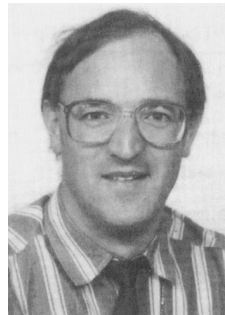
5. **Conclusions**

This paper has briefly introduced the characteristics of cell delay variation within an ATM network and the effect of this on play-out buffer operation and dimensioning. The paper has also shown the effect of cell delay variation on relative clock quality when SRTS or adaptive clock recovery methods are used to synchronise the transmitter and receiver clocks. When the ATM network has to interwork with existing network synchronous platforms such as PSTN and ISDN networks, it is important that the interconnected services are synchronised with the existing platforms using the network synchronous technique to avoid excessive frame slips. Where the ATM network has to transfer third-party timing information between PDH interfaces, the synchronous residual time stamp method is preferred to ensure the exiting bit stream compiles with ITU-T G.823 jitter and wander specifications. For ATM-based services it

may be possible to use an adaptive clock recovery technique which would eliminate the need to deliver a network clock to the customer premises equipment. However, it is important to ensure that cell delay variation caused by network short- and long-term load characteristics and waiting-time jitter is fully accounted for in the design.

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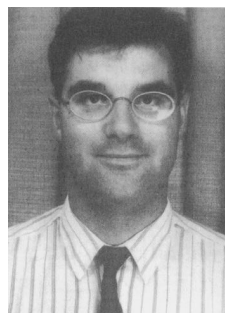


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